ActInf Textbook Group ~ Cohort 6 ~ Session 5 (Chapter 3, Part 1) ~ 2/26/2024

Source: YouTube **Playlist**: Active Inference Textbook Group ~ Cohort 6
Duration: 56:27 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

all right welcome back it's March 4th the only day that's a command March 4th so as we head into chapter 3 does anyone want to begin with any any thoughts or questions or remarks about three or where is it in the context of the book what do they remember from it or what's a quote that was interesting or just a point to jump in from I personally found it really interesting that the marov blanket is just an intermediary it's just a barrier between two things the external and the internal State I really it reminded me of my physics classes talking about a ball rolling down the hill in the sense that you could take it from the perspective of the ball and the universe of moving around it or the the universe is still and the ball is moving down the hill that freedom for that kind of reference surprised me yeah interesting stable Hill moving ball and stable Ball moving Hill that that is that it Zach absolutely yeah yeah that a kind of similar topic came up yesterday in the um discussion with llan Kent and Anna Unica on mental gravity on like kind of there's a reference frame where the Earth like spaceship earth is the egocentric navigation that's like our visual reality like objects coming closer and then there's kind of a third person Sun centrism there's other reference points that can be chosen but that's definitely a very so how would you connect that to markup blanket or to AC or how does the Markov blanket allow us to to do it this way wait I'm sorry perhaps I don't understand are you saying they posed that question or are you asking that question oh just asking like what what made you what made you think about this ball moving down the hill from the markof blanket oh because of the K Divergence it's not exactly perfect but it seems as if you uh can move X the actual like external world is going to stay still but the Q is going to be the thing that's changing however it seems to where your choice as the ball as the thing moving down the hill is kind of arbitrary if you assume that the K Divergence is not going to change if you change up the order it was just a thought that I had that found to be interesting the kind of relative relationship between the two perhaps there's not much to it well well I'm I'm seeing maybe the lrange points as being somewhat porary what are those points so when you have an orbit um when you have two bodies in orbits there are certain points where there's an equilibrium between the gravity Fields between those two bodies so the lrange point is actually a place of stasis um and in a way because it's an absence of both those bodies and the in the influence of both those bodies it's actually a separate virtual body in and of itself so or a virtual Point excuse me that that's very interesting lock Li in in this kind of mental gravity um discussion which maybe people can look at for more detail it's like he did discuss the center of gravity those topics came up with the basian mechanics and Dalton otiva devel like you have a distribution and then you have a center of Mass center of gravity that's like the central tendency of a distribution so basing mechanics is kind of something like a gravitational interpretation of the basian updating process like big distribution heavy prior Light low attention uh sensory input not going to move the prior that much and then the um lrange point is kind of like the the stationarity if you're there like a non-equilibrium stationary Point like uh the person in the frog and the uh jumping out of the hand example a textbook they could reach like a long-term stationerity where they were just estimating something that where they kind of they've reached like a steady point with what their environment was providing so it's kind of like one Gravity from the environment one Gravity from the um agent with their prior and then also that the um p is stable like the true distribution of uh temperature stable Q is changing beliefs are changing or beliefs are stable true distributions are changing and that kind of uh maps to the um change the world change the Mind possibility like belief updating and fixed belief World State changing so nice lot of great comments here Thomas and then yeah yeah two points one is I found really fascinating the Reliance on temporal dimensionality for a lot of the argumentation here so for example the argument of basic the high rical gener generative model for example the illustration of sentences right composed of words uh their high rical orentation is based on a nature of duration things are taking longer than other uh that's a really fat basically things are taking a rapidity a rapidness of movement versus a duration longer period of time um as the argumentation for like what gives an ordering to things I think that's incredibly fascinating another dimension related to this idea of like emotional like why we're talking about this kind of psychological aspect of gravity um there's that line where that idea of temporal depth is directly correlated with psychological dep of a person and that's really interesting on a couple grounds one is I'm attaching this that ideas of the kind of factual and trajectories and policies argumentation that's been presented so far AOS has a really fascinating line in metaphysics about rational action activity he uses a very particular example of a rock a rock is IR R the activity of a rock dropping is not rational because it has it's compelled by necessity to have to go down the point versus rational actions there's AOS a power of Choice a power of D of deral of of either to dox or not do X so the counterfactual point the example use in Ariz metaphysics is uh medicine right to heal or not to heal for example um or that actor basically the agent throwing the rock or not throwing the rock again that cap like that the drawing of rational activity to be that power of choice of temporal optionality basically um I saw some correspondence to notes like that I've always always find that very interesting AOS style he talks about that's his def that's like derivative his explicit definition of what rational activity is uh which has this kind of potentiality temporal nature to it I thought that was really fascinating um the other note it's a classic problem um this is a from a from Scholastic aquinus models of psychology the chasm between the fantasms and the intellectual activity in Scholastic Traditions it's considered a mystery they don't know what's allowing the conversion of sense data into the F What's called the fantasms which I guess in core in this kind of mapping logic would be the active States uh there's no the explanation is kind of obscured how you especially the problem is our sense stum is on particulars a particular dog or a particular expression of light if you're looking at in a room for example how do we do the cognitional action of universals to get the idea of this Quantum is in front of me is the dog there's a whole thing about that in scholas atmology about that what's relevant to this group is I'm a little unconvinced that this argument of the blanket is solving that problem I say that because there what exactly is the C like so if if if there's like three variables that are basically being interchanged with active States external States sense States and internal States you know obviously the correspondence of the variables between active States and external States and sense states to internal States totally get it like I could see how this Maps right uh but between the variable X and Mew what is that c like how is that connected I'm not I didn't see anything in the documents in the literature so in this chapter so far to give that explanation of how that conversion is happening uh I might I may have Mis read that I might completely be a mistake but this is just in my mind I just know this is a traditional uh problem in epistemology of basically the application of universals of perception and cognitional activity there's a no one really knows how to explain give an account for that so that's why this chapter caught my atttention and I'm a little confuzzled about this uh because eventually when he get drawn down to be uh yeah unit U Bas of the variables become U and Y when really mu and X are seem to be the question mark of what that is the thing of of translation uh between the sense internal States and the active external States how is that happening I don't see that happening yet um okay those those are just my generally notes again I might be widely wrong about this so I'm happy to be corrected on that more strongly stated second point but also that first point on specifically the temporal nature of of arrangement of generative hierarchies and synchronicity coming in is a very important factor I saw I read that I actually have that down in my notes um let's see what was that yes that the internal states have an appearance of representing external States so mu dot is representing appropriately x dot by the synchronicity of the active uh state in the sense statey and do that's what I got thank you thank you for the great comments so to the second one first that's a very perceptive comment which is that the um particular partition which is the starting point for a free energy principle active inference and not necessarily starting point to get to any phenomenological or metaphysical Point potentially only as a starting point just to do something empirical so first a failure or a Delta to a philosophical explanation may just be like a permanent open question for a lot of other reasons so the but but just to the kind of philosophical implication part the edges which represent what basically can happen it's basically all the edges can happen except for two kinds of things no backwards action no backwards perception but the other arrows are like in principle double-sided that doesn't mean in any given situation they're like of equal relevance also this is a map not a territory so like just drawing a line here between um you know the the the North and the South side of the city that that that line on the map is like it's literally not describing what the materiality is of the connection so that's going to come back like again and again obviously the map not ter question so first category thing that doesn't happen that reverse sensation I don't have my thumb on the scale of sensation or it's modeled like um part of the sparsity of the models that it just can't happen and the other thing that can't happen is that action can't be controlled externally action is under um one the autonomous control this is not like just pick it up and pce it on to like an agent as complex as a human because as we'll get into like even in a tze we'll see how limited it can be so this is just a starting point of kind of like the the like OnePlus One Lego block that then more sophisticated models especially with internal sophistication RS um but the other kind of thing that doesn't happen other than the backwards action backwards sensation is the no telepathy no no um telekinesis I guess like no no communication from the outside except through sensation no action effect on the ex outside out except for through action so that kind of gets this question of like um how but isn't what we actually care about the coupling between the systems which is like the The Edge that would connect these two I think the um [Music] um it it's not with the Conor hind was 26 um basing Mechanics for stationary proes that kind of comes back to this um lrange point this one talks about the blanket so here's before partitioning the blanket out into um incoming sensory and outgoing action so this is just the the um initial marov blanket concept on an undirected graph so both of these edges are biral so this is kind of the simpler version of um this where the blank kit is broken out into two then um they study this the Locking of the mapping the actual stationarity in the tracking is between the internal and the external state that is one that is the state St it that is the kind of good regulator like stationarity so it captures both the um intervention of the blanket State and the I I'm I just don't remember exactly which paper has the the sigma mapping between maybe someone else remembers but the sigma mapping is is the one that describes um between the internal and the external state so there is like it's like an image or a map and it Maps between the internal and external States but it's a but it's a it's another um projection but this only describes the causal uh connections Jeff yeah I just I just had a thought um that I really haven't seen much in the literature um until very recently about Magneto reception and also we know that we're also sensitive to gravity but I haven't really seen that modeled as a sense that we actively participate in as well so maybe there's something there when you're talking about so-called telekinesis um where where it's actually just a natural phenomenon that that that may be because we know for example Birds right Birds navigate using the Earth's magnetic field we know that um we know that different different types of animals have different sensitivity to gravity fields we know that when for example when we go Cuba Diving we lose our sense of gravity right um so maybe maybe that's just something to to just stick in your your file cabinet I don't know yeah definitely this live stream from yesterday it was a rare Sunday live stream like it it links to neurobiology and a lot of other things and we we we couldn't think of a linguistic example that didn't convey um that wasn't concordant with like a gravitational like metaphor being applied in a certain way Zach uh hi uh so can you say a little bit about so there's no telekinesis yeah but what if control over one specific variable is in like one Markov blanket and another is in another Markov blanket would the separate Markov blankets under some other hierarchical Markov blanket be considered an external state or would those all be inside of one big blanket that you just call the set of internal States if that makes sense yeah um though in this like minimal form they're labeled this way they're all relative to each other and that's all relative to the choice of which one is chosen by the map drawer as the internal state so it's not like aspects of the world are like oh this one's a sensory State now we can move on it's like in this model where this is the internal State and this is the external State then they have this directionality um there's an interesting question I think it's in it's in chapter two or three and it kind of highlights how the uh textbook talks a lot about generative model and generative process but then um it has Alli and then uh but more recently they just described the entire statistical model as the generative model so that's the way that um many people talk about generative modeling now that although there are agents that are defined by like the causal connections of the blanket in the internal States so like you can have agent based model as as you could with net logo or any other kind of modeling framework um there's no need to talk about specific variables as being intrinsically like an active state or a sensory State because if it was like a communication setting then the active for one is the sensory of the other so as soon as you're even in a a two agent setting or anything that's um like blurring the line with the basil cognition or anything you just build the program and you use the structure the syntax of the of the system that you're building the model in like we can look at pmdp and look at the code as it's kind of rendered for some people that is helpful to see that these are really just variables that are defined in their relationships um and then when you start linking them up they're they're they if you had another you know extension it's was a question that that we hav't and I don't know if there's a specific total answer which is like okay it's a wiring diagram like this um and this was the kind of core critique or or line of um inquiry by this like how particular it's like okay the FP has these properties but does it retain those properties like as you change the topology of the ACT ction perception Niche like if you don't have this double edged Arrow or like like if you just had Around the Clock like only one two 3 four what properties does that coupling have can that one synchronize um across blanket uh so I I I don't think it's totally resolved uh but it can be explored how different connectivities and of course different parameterizations within a connectivity relate to like different functions and especially when getting into um internal states that can have um Dynamic structure so you'd say okay the temperature in the room is going to just be like it is what it is and then the um thermometer and the air conditioner kind of also are what they are but we have an agent who can create higher order abstractions about temperature through structure alert so a lot of the change happens like already blanketed off from the sensory environment which I think kind of touches on Thomas's Pro problems questions too awesome thank you so much pretty interesting figure cool visualization could probably be used more broadly or like something like that like some you using different colors and types oh here's the mapping here's Sigma mapping x y and blanket so X and Y and then sense in action let's see past time Point current time point so here's just describing the X maps to action and itself internal States our action selection and have their own recurrent Dynamics and like okay analogously two Sol little coupling um I mean we can explore these more in the paper I don't think this is the same q but I think this reflects that kind of like um kind of like lrange Point like non-equilibrium steady state something about how they're like in a something like they're circulating in some way three is the blanket condition saying it's this set composed of sense and action summarized with this one node and then the assumptions some other kind of conditional Independence like that they all have this one other here they look more connected here's the sigma mapping here's something else about this observations or something pretty cool I don't know how much more graphical but but the category theory is also taking it very graphically this is still within the basian graph like visualization approach but the category Theory work probably has a lot of visualization that that's also more formal argumentation any anyone else just any other angle or we can go to some questions also people can upvote the questions but let's just look at does anyone want to look at a specific written one or new one or should we just look at the most uploaded ones Markov blanket explanation page 43 Box 31 so here's the marov blanket in time concept the or which is sometimes called the marvian property like as has kind of been discussed before the last names don't really add any information because there's like multiple marov and there's multiple senses and all these things so sometimes just using the descriptors more informative but the marvian Assumption like it just broadly in statistics like talking about like a Marian model or hidden marov model is where the past time step influences the next time step and then that influences the future time step so the past only influences the future through this kind of retained One Step present um how can this be true it doesn't have to be a true fact ontologically about a system it could just be a a statistical approximation or designed feature like a program what if we don't know the present well that's a super broad question by even sketching out like a few nodes on a piece of paper or thinking a little bit about a specific system like what like what system not knowing what about the presence just sketching that out moves along that question but the general form is probably pretty big question they Define markco blanket well they Define it locally but it's also prior defined the set of variables that mediate all statistical interactions between a system and environment what's the statistical rather than a non-statistical interaction um here maybe in the broader sense it's saying like it's about the statistics of the map not the actual territory so we're already in the domain of statistics another um way to read it would be like it's kind of a probabilistic like across like many similar or comparable examples that it's not necessarily seeking to give like for example a mechanistic account or constructivist account of how one thing happens it's giving a observational empirical numerical approach to modeling like categories of events just like anyone would doing statistics or AI or machine learning Thomas kind blending questions one and two together and what's the relationship of hidden states with this because I remember the model from the you know low road Mo chapter there was a discussion about hidden States being part of the basically the calculation um so just that question of not obviously the point of not having all full information is covered by the fact that there's hidden states that are being accounted for uh as I recall that from the chap on chapter two in h in the low road process so is that part of that relationship here one of the ways to rec the connect the two I questions yeah like beliefs about hidden states do come into play in the variational free energy and then also in prospective planning and action selection like in this book anytime you see X till the um that's beliefs about hidden states are always um more uncertain than beliefs about measurements so these two forms of expected free energy um expect like also always good to just kind of like threshold that not all things necessarily do this kind of policy select ction process Like a Rock doesn't have to be modeled with a policy selection process or just a trivial one but then getting into this like adaptive action setting then here's like the uncertainty about how actions relate to World States is always greater than uncertainty about how actions map onto observations best Cas they're equivalent like in chess where there's a one toone mapping between the um observation and the hidden state of the piece on the board that's like a fully observable game not to say that the opponent's mind is fully observable like but that's the whole modeling um process uh those are maybe a few aspects of it but what else what what what are the kind of key aspects you think and how maybe it's been approached philosophically otherwise like oh are you ask or just like stat yeah kind of this is a statistical model but not all thinking on this issue has been statistical so how just as kind of open question but maybe if you have any thoughts of any pre- statistical thinkers and how they approach like boundary I mean we normally argumented by definition so the idea of an Essence is defined by definition which is a speci in general with the differential basically combination um if you use mean Plato in Theo talks about that using NE necess necessary con property to Define limit and thus used it to Define color and figure and all that um so those are non-statistical probabilities to define basically some observation um in fact actually from a more Norm like from a more ethics normative point of view uh the idea of examplars is the basically exact opposite that the best representation of reality is the examplar versus uh an statistical average right um so that would be kind of the broadest kind of answer I've agreed that broadest difference between a statistical model and a non-statistical is just use of definition and Essence uh from a CL from a traditional classic point of view cool really cool yeah that's awesome thank you this is like the N equals one and the example and the a priori non-empirical and then the statistical like or then the instrumental would kind of which is just described basically using it as a statistical model like people often talk about this this is the whole not the whole but the a lot of the discussion around like do we do the organism are they a generative model do they have one are we just modeling it as such that's this instrumentalism realism question I mean if known claims are being made about the territory then you know one can always Retreat to just pure map making but then there are stronger guarantees of the once you've retreated to map making but then that has fundamentally given up certain territory relating to like how individual things happen it's just and it's just a different kind of it's a different kind approach and and I I I know there's some connections with the free frequentist and the basian statistics and like how do frequentists view chance how does basian view chance there's probably a lot to go in there too yeah just to build on that just I just by the way it wasn't like Aristotle did not know about chance it was discussed it's a know it was a known thing the I think the big difference just to CL I mean that's drawing on this point is that science ofis was supposed to be about universals like this idea of kind of use of chance to kind of observe and find reality is that's the part where there probably a huge Schism difference um just from ran Point uh epistemology or investigation there something different so it was like they know they knew chance existed and they definitely acknowledged that certain things do apply to that but it wouldn't be part of a science it wouldn't be the mo the way you would use to come to find the universals is through random chance yeah yeah and the role of abduction and induction deduction and kind of which ones are given what epistemic statuses does anyone want to go to any written question or or any new one here's no answer but this lran again that's the last the last names points like it's it's hard to know is this lrange point it could be but it might be totally unrelated it could be different person different work or it could be exactly the same but like a point attractor or something like that gives more information okay let's do let's just see how many from the top trying to understand better how figure 3.2 was produced EX exactly what it means okay so there are the variables that are going to be used this is kind of the simple blanket setting there's going to be three variables X mu and B so this is just in the undirected graph setting before splitting out sense and which is what kind of transposed it into the cybernetics context but just one blanket State um and then these are kind of three projections down there's a cube of statistical outcomes in the um like variables taking on a number in the X Y and Z Direction but it's in the X B and mu directions and then here are three projections from the cube on each of the faces down onto each faces um just comparing two different variables um so uh there's a positive relationship between just a linear aggression between X and B and then you can condition on B like you could kind of zero out like you could talk about how many um standard deviations each X was away from the conditioned expectation on B so like this one would be or this one would be lower in standard deviations than this one just be it's like the residual from the regression so there's just statistical methods and there's other relationships that are drawn so these just describe the like conditional dependencies like kind of this this is like a statistical Edge like but that is drawn that is saying the scatter plot has a a significant explanatory power um but it's not exactly tested like with a P value as such but just saying that that's the slope is like the the causal effect if the causal Edge were one then all the points would just be on a line like knowing one would be like 100% like which is one in statistic everything's kind of scaled to one everything happens on the 0 to one interval Eric or Thomas and anyone else got to follow up on basically my my question about what what brings the what pulls the X and mu value variable together like this grap seems actually I had that in my notes I forgot talk about that right because what happens it looks like it's using the blanket B to that's the that third graph on the bottom left is the combination right of the two V of those two investigations or two measurements and then that fourth graph is the probability mapping of x given uh look at the note yeah Pro given the expected probability of mu given B conditioned on mu I think best understand to read it yeah great that that that adds a lot definitely so like these two describe the conditional dependencies between X and mu so internal external States on blanket and then here is that mapping so it's more variable like let's just say that there's a 08 relationship between the temperature in the room and the thermometer noisy thermometer 80% good thermometer and then um there's a 8 Fidelity between me looking at the numbers on the thermometer so just coarsely it's like then there's going to be a 64 relationship Sigma de facto like around horn because of correlations amongst variables so yet there's no direct causal effect between the hidden State and the the external State and the internal state so that that describes both like the the intermediation and the kind of ultimate correlation and so this is this is like structural equation modeling is commonly used like in population health or the um earlier or it's oh it's in chapter four here's like some some kind of here's one hidden factor causing one observation here's one here's lightning and then here's the light part and the sound part two observations from one underlying State here's two causal factors that combine to influence one outcome these kinds of Bas and Gra iCal methods are used like from like knowledge database systems to health modeling all these different kinds of things okay if one defines these preferred States as expected States then one can say the living organisms must minimize the surprise of of their sensory observations is it that preference is the same as expectation or formally we can use them in the same equation or something else does active inference make ontological claims EG preference is actually expectation just as heat is actually exitation of molecules more generally is a free energy principle free energy in physics just an analogy or is an ontological than okay one important thing to note is that there's two senses of expectation so in everyday usage the meaning or the connotation is like what we expect to happen like what what what we expect will be the weather tomorrow however also when we see the fancy e the Big E expectation of or the sum of something divided by n data points which same as the the that's also called the expectation also known as like the central tendency or the mean so the expectation of a distribution it's like the expected returns but that's a situation that's both about an uncertain future moment and it has to do with the moment with the central um tendency so um it like it's just a statist even within statistics those are just clear based on the context or when what variable um is coming to play um inactive inference preferences so four things that are choosing actions that are using their that are exerting their preferences let's look to chapter seven so here's a model where there's no preferences involved that helped calibrate some of these statistical questions about like the prior the relationship between the hidden State inference and the observations through time through the a matrix and like the B Matrix in how the hidden state is inferred to change through T but there's no action exerted in this model but that brings out a lot of these questions about like observation hidden State how things change in the world all these things like that then that bottom structure is still exactly the same and then here's this action selection part that invokes this variable C which is called preference it's not a reward function it's a preference function that's a distribution over observations so it'll again come up in um like chapter 7 and before what what its structure is but this is one of the core pieces of active inference models that the preference component of the selection which is going to come into play in expected free energy in the pragmatic value so this is going to Define pragmatic value for that variable that is defined in terms of observations sensory observations let's return return to the question see if there was more to it but that's kind of the key pece they're not that preferences are not about future points preferences come into they're not using that future um unknown outcome sense of expectation but you could imagine a planning agent that also has preferences about future States or their preferences right now C right now comes into play in deciding action that matters for future events but that's preferences over sensory outcomes now having influence on action selection not like a betting Market on what an observation is going to be at a future time Point per se so is preference the same as expectation no but it'll be especially clear when you just even sketch out a table or a list of what is being described because this is a very general this is like at the level of definitions of words formally we can use them in the same equation there are situations where it's an equation where you could describe it in English using either of those words but there's other situations where you you couldn't so they're not simply swappable for that reason this can't be really described also I don't I don't know about this heat is actually excitation molecules kind of interesting metaphor but maybe write more about it and then let's explore it is free energy in physics just an analogy or is an ontological assertion can analogies be ontological assertions I don't know there's probably a lot of things say on that question but we'll go on what's the difference between the terms or does anyone have another in like 10 10 minutes or we can just look up like a few more of these or anyone can ask or say anything else okay what's the difference between the terms uh Thomas go for it hey sorry I mean the the idea of the generative model hierarchical generative models based on time could we clarify that a little bit more because it does sound like um the fact of a hierarchy right basically for example again the classical model of a of a chain of being that thing that uh entities have a certain orological reality or hierarchy right that rocks are not the same as animals animals are not the same as humans humans and God U there's a kind of ontological hierarchy given on that um of some orientation of how things are arranged this chapter seems to talk about the time duration as that ordering property could you get more clarity that or just like how strong of a statement is that or what the implications are are that yeah greate question okay few pieces on this so um when you brought up the first time let's just say that we're going to have planning over 100 years one kind of model would be like a model where every year from 1 to 100 is represented like on a timeline or there could be a model where there's 10 decades and then Each decade has 10 points and so it's kind of like you could have um th thinner models that are nested structurally so that like the tree sweeps out all the possibilities still and so there's multiple different kinds of like hierarchical models that could be built um you could have you know 11s within a 20 within a 30 is like that doesn't make sense maybe it's not the most statistically optimal but that's that's not outside the space of statistical models that you can build because these are just statistical models however also people have gone I think a lot further in arguing for like time phenomenology um an example here's from 2018 an interview with Carl friston from Alias um with Martin forer so there's a question about that temporal depth so we ask basically well you're highlighting temporal depth a lot like whether like thickness through time so is it enough for a computer program to do it and then if an ant colony is able to persist over longer time scales than other things like it then doesn't it have more thickness to um it returns to this kind of he brings up a lot of different components and I think it it like kind of remains to a very careful analysis like which which claims he he brings out to respond to it um because it includes like questions about meta awareness the depth of the model in terms of the counterfactuals like for example like um use economic model the idea of of time discounting for example and financial modeling the idea that a $100 today has more value than that $100 10 years out because of properties like inflation for example um so yeah absolutely there's that kind of self-referencing quality like kind of Agents having self-awareness question but also this this temporal time domain has kind of external reality qualities if we consider time preferencing uh of value itself being a variable here or being an area of consideration yeah I think the deeper um the the the deeper philosophy is to be explored here's what it looks like in the code policy length equals four that's the time depth of that agent for time steps like that lets you estimate how many um how big the policy Vector is going to be if there's four times uh if there's four um affordance is or let's say five affordances like up down left right and weight at each time step then policy length of four is just every sequence of those five affordances so five to the 4th and then that's like tree search for policy just like early chess algorithms and then there's all these methods that that are that are different but the starting point is just like the model defines the temporal depth of the plan now what does this say about the the actuality of time that's probably pretty open thank you yeah thank thank you for the question yeah and and then like that is what motivates all these okay well now how about inference on policy length and all these other things like it's just about starting with a model iterating with it seeing what phenomena of the real system still AR are unaccounted that that gets far okay in last minutes any questions okay let's see so that was the first discussion yeah Eric I'm sorry uh this this might be a little straightforward to some but whenever we were talking about x dot whenever we were talking about the Dynamics of internal external as well as boundary state the dot did that mean with respect to the derivative of the actual boundary uh like variables or was that with respect to time now now I'm kind of questioning myself great question if it's no explain it explain oh I was just going to say if that would take us over time we can uh discuss it afterward in the uh questions yes I I think at a first reading the dot on these are the the derivative of the variable so that the value at when it's zero that is a stationerity of that variable so but the Dynamics of the variable are defined in terms of a function of those other states plus a noise parameter which is what makes it amenable to like all these kind of stochastic calculus physics approaches because this like separation into a signal in a noise term on a flow is the basis of a lot of other formalisms and it's always hard to also I think jump between this is like the physics formalism like physics of particular systems flows L Gran all of that but then when we're actually doing the modeling in discret setting it's like where's the marov blanket here because here it's just a partially observable marov decision process but people have had that for decades so clearly they didn't need the particular physics to do a PDP so that's kind of an interesting thing to explore too okay next week at the later time second part of three otherwise in the other cohort we're in chapter eight which is continuous time models but seven is on discreet time so thank you all see you next time bye